R1.3 Energy from fuels

Fuel cells

Learning outcomes

By the end of this section you should be able to:

- Compare and contrast hydrogen and methanol fuel cells.
- Deduce half-equations for the electrode reactions in a fuel cell.

Around the globe, many countries are turning to renewable and/or green energy sources to try to solve the world's energy needs, while still addressing our ever-increasing concerns about the greenhouse effect and global climate change. Some countries have the natural resources and infrastructure necessary to make this transition, but other countries do not. In some cases, countries still rely heavily on the combustion of fossil fuels and biofuels to generate electricity. But what other ways are there to derive energy from molecules? Is there any way to avoid the combustion of fuels to meet our energy requirements?

Fuel cells vs. conventional techniques

Electricity can be produced from the combustion of fossil fuels such as coal and natural gas. In these power plants, the heat released from the combustion reaction undergoes many transfers and transformations to become electrical energy. In a fuel cell, electrical energy is derived directly from a chemical reaction without the need for multiple transformations of energy. As such, fuel cells are highly energy efficient compared to their classic counterparts.

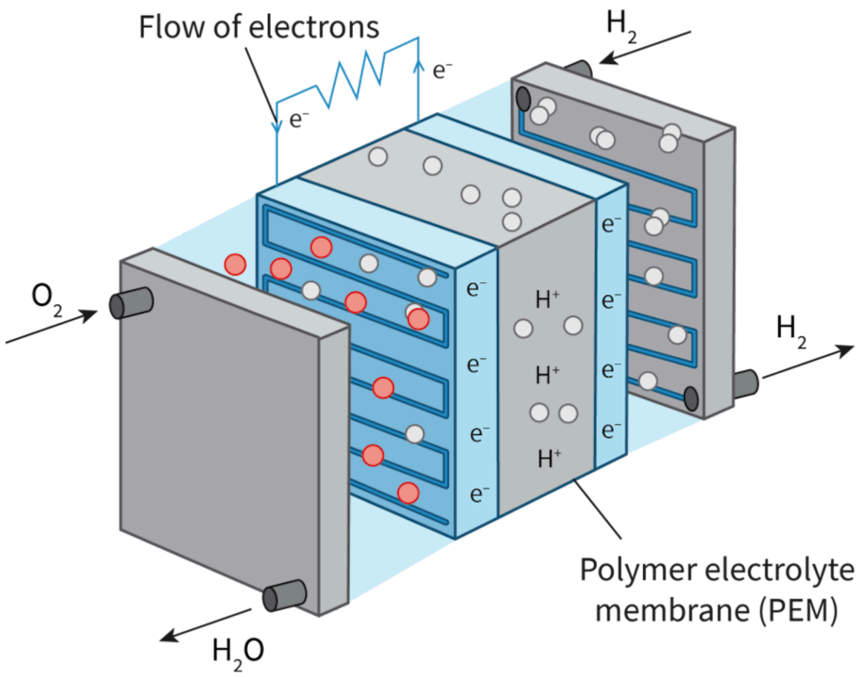

Figure 1. A simple fuel cell.

More information for figure 1

The diagram shows a simple fuel cell that includes several key components and labels. On the left, oxygen (O2) is depicted entering the cell. On the right, hydrogen (H2) is shown entering from two directions. Water (H2O) is indicated as a product flowing out.

The central part of the diagram consists of a polymer electrolyte membrane (PEM), which separates the cell into two regions. These regions are connected by a pathway labeled as the flow of electrons (e-), representing the movement of electrons from one side of the membrane to the other.

The diagram illustrates ions and electrons participating in the reactions. H2 splits into H+ ions and electrons (e-) at one electrode. These electrons travel through an external circuit path, indicated by a zigzag line labeled "Flow of electrons," to provide electrical power.

Hydrogen ions (H+) move through the PEM to meet oxygen at the other electrode, forming water (H2O) as a byproduct. The entire process illustrates the conversion of chemical energy to electrical energy in a fuel cell.

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A fuel cell consists of a spontaneous exothermic chemical reaction that is divided in two regions. These two half-reactions occur separately in two different locations, separated by a membrane, but connected by a wire. The wire allows for the flow of electrons from one half of the reaction to the other.

Electrons are formed at the side of the chemical reaction where oxidation is occurring. The electrons flow into the wire and then towards the other side of the fuel cell where they are used. The movement of electrons through the wire creates an electrical current that can be used to directly charge a load or operate a piece of machinery.

Methanol fuel cells

Let us take as an example a combustion reaction that you studied in section R1.3.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/combustion-reactions-id-43476/). Here we are combusting methanol in the presence of oxygen gas to produce carbon dioxide and water vapour.

$$\mathrm{CH_3OH(g)} + \tfrac{3}{2}\mathrm{O_2(g)} \rightarrow \mathrm{CO_2(g)} + 2\mathrm{H_2O(g)} \quad \Delta H = -711\ \mathrm{kJ\ mol^{-1}}$$

A methanol fuel cell works with the same chemical reaction but separates the process into two half-reactions.

Half reaction:

$$\mathrm{CH_3OH(g)} + \mathrm{H_2O(l)} \rightarrow \mathrm{CO_2(g)} + 6\mathrm{H^+(aq)} + 6\mathrm{e^-} \quad \text{loss of electrons}$$

Half reaction:

$$\tfrac{3}{2}\mathrm{O_2(g)} + 6\mathrm{H^+(aq)} + 6\mathrm{e^-} \rightarrow 3\mathrm{H_2O(l)} \quad \text{gain of electrons}$$

Adding these two half-reactions together produces the following net-reaction, which you can see is the same as the combustion of methanol with reduced coefficients. You may have noticed that combustion reactions are sometimes balanced using fractions as coefficients in order to make sure that the combustion reaction is written per mole of fuel combusted. This allows for a better comparison of energy produced per mole of fuel. In addition, the balanced overall reaction must be equal to the sum of the two half-reactions, including the correct states of matter and molar coefficients.

$$\mathrm{CH_3OH(g)} + \tfrac{3}{2}\mathrm{O_2(g)} \rightarrow \mathrm{CO_2(g)} + 2\mathrm{H_2O(g)} \quad \Delta H = -711\ \mathrm{kJ}$$

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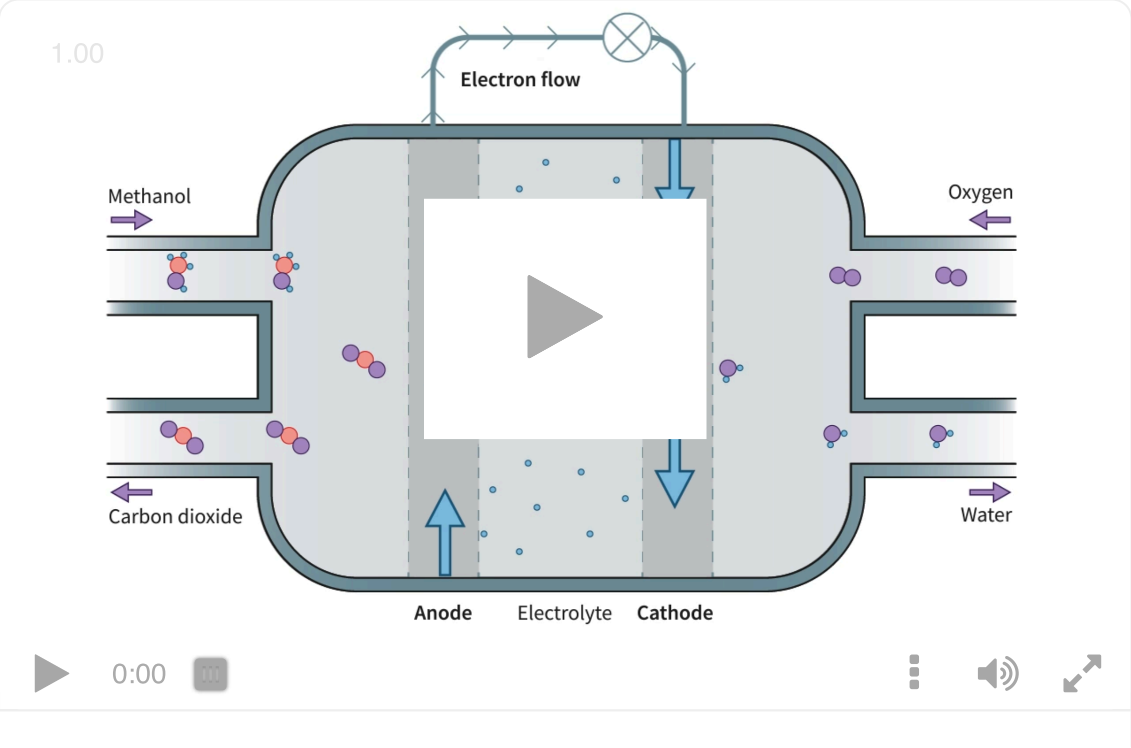
When electrons are lost by an element the element is said to have undergone oxidation. When electrons are gained by an element the element is said to have been reduced. These concepts will be explored in more detail in section R3.2.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/oxidation-and-reduction-id-43487/).

Use **Interactive 1** to explore how a methanol fuel cell functions.



Interactive 1. Functioning of a Methanol Fuel Cell.

More information for interactive 1

This diagram shows how a methanol fuel cell generates electricity by converting chemical energy into electrical energy through a redox reaction.
On the left side, methanol (CH_3OH) and water (H_2O) enter the fuel cell through an inlet. A chemical reaction occurs here where methanol is oxidised, releasing carbon dioxide (CO_2), hydrogen ions (H^+), and electrons (e^-).
The hydrogen ions move through a special membrane in the centre of the cell, while the electrons travel through an external wire, creating an electric current that powers the bulb or the device.
On the right side, oxygen gas (O_2) enters through another inlet. Here, the oxygen combines with the hydrogen ions and electrons, forming water (H_2O) in a reduction reaction. The flow of electric current is indicated by the continuous flow of an upward arrow labelled anode and the continuous flow of a downward arrow labelled cathode. On the left, water is emitted out through the outlet. This setup is an efficient way to produce electricity, with only carbon dioxide and water as byproducts.
This interactive explains how a methanol fuel cell works using the concepts of oxidation, reduction, and ion flow. Identify the inputs (methanol, water, oxygen) and outputs (carbon dioxide, water, electrons) of the reaction.

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It is interesting to note that fuel cells are supplied with a continuous flow of reactant molecules, which are pumped into the cell. Unlike stand-alone batteries, fuel cells do not need to be charged over long periods. Methanol fuel cells can be refilled from a liquid methanol reservoir, which makes them convenient to use.

Hydrogen fuel cells

Since the advent of space exploration NASA and other agencies have been using hydrogen fuel cells as a portable method of powering the electrical systems on board spacecraft. A similar process occurs in hydrogen fuel cells compared to methanol fuel cells. In this case, the spontaneous and exothermic reaction between hydrogen gas and oxygen gas to form water is harnessed. This reaction releases approximately 286 kJ of energy per mole of hydrogen gas reacted.

Let us examine the fuel cell under acidic conditions.

Where the net reaction:

$$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$$

is divided into the following two half-reactions.

Half reaction:

$$2H_2(g) \rightarrow 4H^+(aq) + 4e^- \quad \text{loss of electrons}$$

Half reaction:

$$O_2(g) + 4H^+(aq) + 4e^- \rightarrow 2H_2O(l) \quad \text{gain of electrons}$$

Making connections

As seen in section S3.1.6 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/oxidation-states-id-44696/). At the anode, the hydrogen atoms in hydrogen gas have an oxidation state of zero, yet as hydrogen ions, they have an oxidation state of +1. This change in oxidation state from 0 to +1 involves the loss of electrons. Loss of electrons is said to occur at the anode.

On the other hand, at the cathode, the oxygen atoms in oxygen gas have an oxidation state of zero, yet as oxygen in water molecules, they have an oxidation state of −2. This change in oxidation state from 0 to −2 involves the gain of electrons. The gain of electrons is said to occur at the cathode.

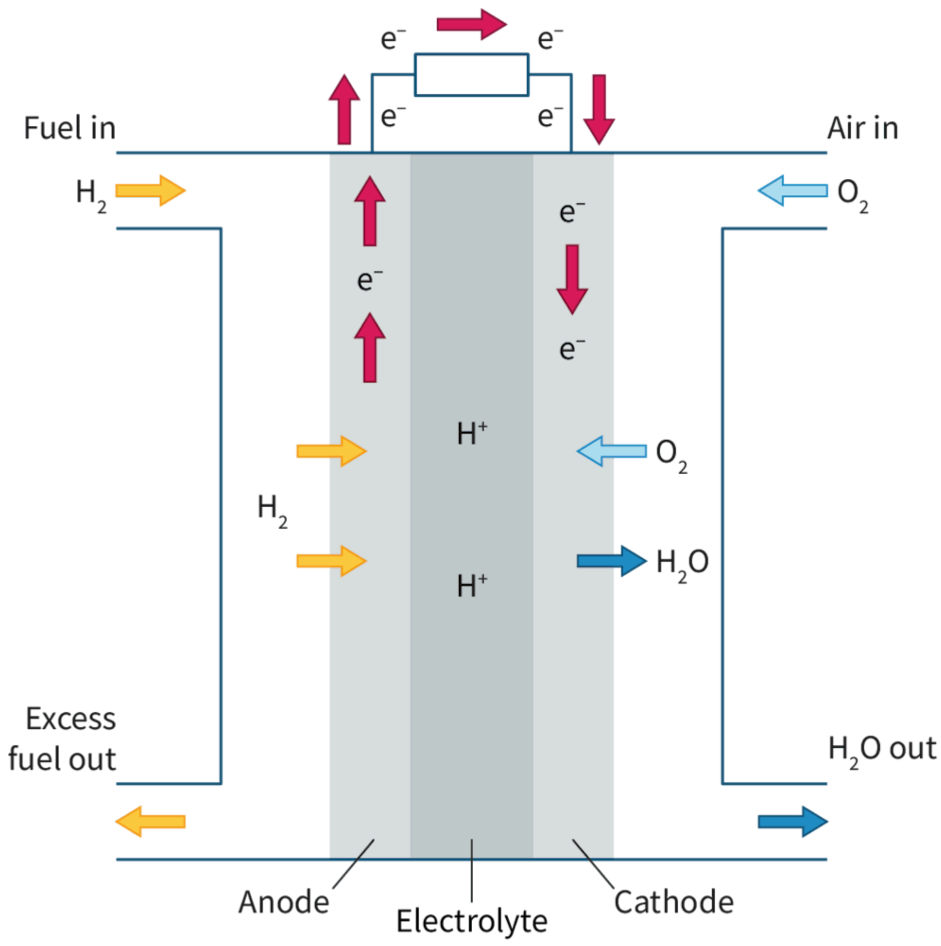

Figure 2. A hydrogen fuel cell.

More information for figure 2

The diagram illustrates a hydrogen fuel cell. It comprises three main sections: anode, cathode, and the central electrolyte area where the reaction occurs. On the left, labeled "Fuel in," hydrogen gas enters the anode. On the right, "Air in" is labeled, showing oxygen entering the cathode. Water is labeled as exiting at the bottom, "H2O out." Inside the cell, arrows indicate directional flow; upward at the anode and downward at the cathode. A symbol for electron flow moves from the anode to the cathode above the main cell body, indicating the generation of electricity.

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Though hydrogen fuel cells produce only water as products, it is always important to consider the overall reaction when assessing its environmental impact. It is energetically very difficult to obtain hydrogen gas, as most hydrogen gas on Earth is trapped in the form of water. To free the hydrogen in water it must go through a process known as electrolysis, which requires a tremendous amount of energy.

Hydrogen gas is also produced by a chemical process known as coal gasification, where coal is heated in the presence of steam to produce hydrogen gas and carbon monoxide gas. This method of producing hydrogen gas yields the same atmospheric pollutant as the incomplete combustion of fossil fuels and is therefore not environmentally sustainable. The process also requires a considerable amount of energy as the reaction takes place at high temperatures and pressures.

Activity

- **IB learner profile attribute:**
  - Knowledgeable
  - Communicator
- **Approaches to learning:**
  - Communication skills — Applying interpretive techniques to different forms of media
  - Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 20—30 minutes
- **Activity type:** Individual activity

Instructions

1. Study this interactive: https://sepup.lawrencehallofscience.org/sepup-fuel-cell-simulation/ (https://sepup.lawrencehallofscience.org/sepup-fuel-cell-simulation/)
2. Using the interactive:

   (a) State the half-reaction that would occur at the anode in a hydrogen fuel cell.
   (b) State the half-reaction that would occur at the cathode in a hydrogen fuel cell.
   (c) List the products in the overall reaction in a hydrogen fuel cell.
   (d) Describe if the overall reaction absorbs or releases energy.

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(e) Compare the enthalpy of the products with the enthalpy of reactants in the overall hydrogen fuel cell reaction.